HARD · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

# Problem-Solving and Data Analysis

07

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

# Q1

A

**A.** 0.39

Choice A is correct. It's given that $1 \text{ meter} = 3.28 \text{ feet}$. It follows that $1^2 \text{ square meter} = 3.28^2 \text{ square feet}$, or $1 \text{ square meter} = 10.7584 \text{ square feet}$. Since $1 \text{ hour} = 60 \text{ minutes}$, it follows that 250 square feet per hour is equivalent to $\frac{250 \text{ square feet}}{1 \text{ hour}} \frac{1 \text{ square meter}}{10.7584 \text{ square feet}} \frac{1 \text{ hour}}{60 \text{ minutes}}$, or $\frac{250 \text{ square meters}}{645.504 \text{ minutes}}$, which is approximately 0.3873 square meters per minute. Of the given choices, 0.39 is closest to 0.3873.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

# Q2

D

**D.** $y = -1.674x^2 + 19.76x + 745.73$

Choice D is correct. The data in the scatterplot roughly fall in the shape of a downward-opening parabola; therefore, the coefficient for the $x^2$ term must be negative. Based on the location of the data points, the y-intercept of the parabola should be somewhere between 740 and 760. Therefore, of the equations given, the best model is $y = -1.674x^2 + 19.76x + 745.73$.

Choices A and C are incorrect. The positive coefficient of the $x^2$ term means that these equations each define upward-opening parabolas, whereas a parabola that fits the data in the scatterplot must open downward. Choice B is incorrect because it defines a parabola with a y-intercept that has a negative y-coordinate, whereas a parabola that fits the data in the scatterplot must have a y-intercept with a positive y-coordinate.

## Q3

**1.8**

Accept also: 9/5

The correct answer is $1.8$. It’s given that the regular price of a shirt at a store is $\$\,11.70$, and the sale price of the shirt is $80\%$ less than the regular price. It follows that the sale price of the shirt is $\$\,11.701 - \frac{80}{100}$, or $\$\,11.701 - 0.8$, which is equivalent to $\$\,2.34$. It’s also given that the sale price of the shirt is $30\%$ greater than the store’s cost for the shirt. Let $x$ represent the store’s cost for the shirt. It follows that $2.34 = 1 + \frac{30}{100}x$, or $2.34 = 1.3x$. Dividing both sides of this equation by $1.3$ yields $x = 1.80$. Therefore, the store’s cost, in dollars, for the shirt is $1.80$. Note that 1.8 and 9/5 are examples of ways to enter a correct answer.

## Q4

**A**

A.

| Value | Frequency |
|---|---|
| 70 | 4 |
| 80 | 5 |
| 90 | 6 |
| 100 | 7 |

Choice A is correct. The tables in choices B, C, and D each represent a data set where the values $80$ and $90$ have the same frequency and the values $70$ and $100$ have the same frequency. It follows that each of these data sets is symmetric around the value halfway between 80 and 90, or $85$. When a data set is symmetric around a value, that value is the mean of the data set. Therefore, the data sets represented by the tables in choices B, C, and D each have a mean of $85$. The table in choice A represents a data set where the value $90$ has a greater frequency than the value 80 and the value 100 has a greater frequency than the value 70. It follows that this data set has a mean greater than $85$. Therefore, of the given choices, choice A represents the data set with the greatest mean.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

# Q5 B

**B.** 16.3

Choice B is correct. It's given that the speed of a vehicle is increasing at a rate of $7.3$ meters per second squared. It's given to use $1 \text{ mile} = 1{,}609 \text{ meters}$. There are 60 seconds in 1 minute; therefore, $60^2$ or $3{,}600$ seconds squared is equal to $1$ minute squared. It follows that the rate of $7.3$ meters per second squared is equivalent to $\frac{7.3 \text{ meters}}{1 \text{ second squared}} \frac{1 \text{ mile}}{1{,}609 \text{ meters}} \frac{3{,}600 \text{ seconds squared}}{1 \text{ minute squared}}$, or approximately $16.33$ miles per minute squared. The rate, in miles per minute squared, rounded to the nearest tenth is $16.3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

# Q6 C

**C.** 94.25%

Choice C is correct. It's given that $37\%$ of the items in a box are green. Let $x$ represent the total number of items in the box. It follows that $\frac{37}{100}x$, or $0.37x$, items in the box are green. It's also given that of those, $37\%$ are also rectangular. Therefore, $\frac{37}{100}0.37x$, or $0.1369x$, items in the box are green rectangular items. It's also given that of the green rectangular items, $42\%$ are also metal. Therefore, $\frac{42}{100}0.1369x$, or $0.057498x$, items in the box are rectangular green metal items. The number of the items in the box that are not rectangular green metal items is the total number of items in the box minus the number of rectangular green metal items in the box. Therefore, the number of items in the box that are not rectangular green metal items is $x - 0.057498x$, or $0.942502x$. The percentage of items in the box that are not rectangular green metal items is the percentage that $0.942502x$ is of $x$. If $p\%$ represents this percentage, the value of $p$ is $100\frac{0.942502x}{x}$, or $94.2502$. Of the given choices, $94.25\%$ is closest to the percentage of items in the box that are not rectangular green metal items.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q7 B

**B.** 90

Choice B is correct. The mean of a data set is calculated by dividing the sum of the values in the data set by the number of values in the data set. It's given that the mean length of the 240 gray seals from Muskeget Island was 88 inches. This can be represented by the equation $\frac{x}{240} = 88$, where $x$ represents the sum of the lengths, in inches, of the 240 gray seals from Muskeget Island. Multiplying both sides of this equation by 240 yields $x = (240)(88)$, or $x = 21{,}120$ inches. It's also given that the mean length of the 120 gray seals from Sable Island was 94 inches. This can be represented by the equation $\frac{y}{120} = 94$, where $y$ represents the sum of the lengths, in inches, of the 120 gray seals from Sable Island. Multiplying both sides of this equation by 120 yields $y = (120)(94)$, or $y = 11{,}280$ inches. Therefore, the sum of the lengths, in inches, of all 360 gray seals is $21{,}120 + 11{,}280$, or 32,400. Dividing this sum by 360 yields $\frac{32{,}400}{360}$, or 90 inches. Therefore, the mean length of all 360 gray seals the scientist measured for this study is 90 inches.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the mean length of 88 inches and 94 inches, not the mean length of all 360 gray seals the scientist measured for this study.

Choice D is incorrect and may result from conceptual or calculation errors.

# Q8

**A**

**A.** 588

Choice A is correct. It’s given that the sample is in the shape of a cube with edge lengths of $0.9$ meters. Therefore, the volume of the sample is $0.90^3$, or $0.729$, cubic meters. It’s also given that the sample has a density of $807$ kilograms per $1$ cubic meter. Therefore, the mass of this sample is $0.729$ cubic meters$\frac{807 \text{ kilograms}}{1 \text{ cubic meter}}$, or $588.303$ kilograms. Rounding this mass to the nearest whole number gives $588$ kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is $588$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

# Q9

**.48**

Accept also: 12/25

The correct answer is $.48$ . It's given that the number $a$ is $70\%$ less than the positive number $b$. Therefore, $a = 1 - \frac{70}{100}b$, which is equivalent to $a = 1 - 0.70b$, or $a = 0.30b$. It's also given that the number $c$ is $60\%$ greater than $a$. Therefore, $c = 1 + \frac{60}{100}a$, which is equivalent to $c = 1 + 0.60a$, or $c = 1.60a$. Since $a = 0.30b$, substituting $0.30b$ for $a$ in the equation $c = 1.60a$ yields $c = 1.600.30b$, or $c = 0.48b$. Thus, $c$ is $0.48$ times $b$. Note that .48 and 12/25 are examples of ways to enter a correct answer.

## Q10 2

The correct answer is 2.6. Since the mean of a set of numbers can be found by adding the numbers together and dividing by how many numbers there are in the set, the mean mass, in kilograms, of the rocks Andrew collected is $\frac{2.4+2.5+3.6+3.1+2.5+2.7}{6}=\frac{16.8}{6}$, or 2.8.

Since the mean mass of the rocks Maria collected is 0.1 kilogram greater than the mean mass of rocks Andrew collected, the mean mass of the rocks Maria collected is $2.8+0.1=2.9$ kilograms. The value of x can be found by writing an equation for finding the mean: $\frac{x+3.1+2.7+2.9+3.3+2.8}{6}=2.9$. Solving this equation gives $x=2.6$. Note that 2.6 and 13/5 are examples of ways to enter a correct answer.

### Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

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